Part 1

1.1 $\Pi_{Fname, Lname}(\sigma_{Address = \text{"Bangkok"}}(Employee))$

1.2 $\Pi_{Fname, Lname}(\sigma_{Salary > 50,000}(Employee))$

1.3 $\Pi_{Dname}(Department \bowtie__(Department.Dnumber = Project.Dnum \wedge Project.Plocation = \text{"Bangkok"})\ Project)$

1.4 $\Pi_{Fname}(Employee \bowtie_(Employee.Ssn = Works_on.Essn \wedge Works_on.Hours > 20)\ Works_on)$

1.5 $\Pi_{Fname}(Employer \bowtie_(Employee.Ssn = Dependent.Essn)\ \rho_{X(Ssn)}(\Pi_{MgrSsn}(Department) \cap \rho_{X(Ssn)}\ \Pi_{Essn}(Dependent)))$

1.6 $\Pi_{Pname}(Project) - \Pi_{Pname}(Project \bowtie Project.Pnumber = Pno(Works_on \bowtie Works_on.Essn = Ssn\ (\Pi_{Ssn}(Department \bowtie Department.Dnumber = Dno \wedge Department.Dname = \text{"Research"}(Employee)))))$

Part 2

2.1 Database Schema design

Table: Doctor

Primary Key: DoctorID

Attribute	Data Type	Key	Description
DoctorID	VARCHAR(10)	PK	Unique identifier for the doctor.
DoctorName	VARCHAR(100)	-	Full name of the doctor.
Specialty	VARCHAR(50)	-	Medical specialty, e.g. Cardiology.
Phone	VARCHAR(15)	-	Doctor's contact phone number.
YearsExperience	INT	-	Number of years of medical practice.

Table: Patient

Primary Key: PatientID

Attribute	Data Type	Key	Description
PatientID	VARCHAR(10)	PK	Unique identifier for the patient.
PatientName	VARCHAR(100)	-	Full name of the patient.
Gender	CHAR(1)	-	Patient's gender (M/F).
BirthDate	DATE	-	Patient's date of birth.
Phone	VARCHAR(15)	-	Patient's contact phone number.

Table: ExaminationRoom

Primary Key: RoomID

Attribute	Data Type	Key	Description
RoomID	VARCHAR(10)	PK	Unique identifier for the examination room.
RoomType	VARCHAR(50)	-	Type/purpose of the room, e.g. Cardiac.
Floor	INT	-	Floor number where the room is located.
Capacity	INT	-	Maximum number of people allowed.
Status	VARCHAR(20)	-	Current status, e.g. Available/Occupied.

Table: Appointment

Primary Key: AppointmentID (composite candidate: DoctorID + AppointmentDate + AppointmentTime, since one doctor cannot have two overlapping appointments)

Attribute	Data Type	Key	Description
AppointmentID	VARCHAR(10)	PK	Unique identifier for the appointment.
PatientID	VARCHAR(10)	FK -> Patient.PatientID	The patient being seen.
DoctorID	VARCHAR(10)	FK -> Doctor.DoctorID	The doctor conducting the appointment.
RoomID	VARCHAR(10)	FK -> Examination Room.Room ID	The room used for the appointment.
AppointmentDate	DATE	-	Date of the appointment.
AppointmentTime	TIME	-	Time of the appointment.

Relationships

Doctor (1) ──< Appointment (M) >── (1) Patient — one doctor has many appointments; one patient has many appointments.

ExaminationRoom (1) ──< Appointment (M) — one room can be used for many appointments (at different times).

Doctor

Super keys : {DoctorID}, {DoctorID, DoctorName}

Candidate key : {DoctorID}

Primary key : DoctorID

Reason : short, unique and doesn't change overtime.

Patient

Super keys : {PatientID}, {PatientID, PatientName}

Candidate key : {PatientID}

Primary key : PatientID

Reason : short, unique and doesn't change with Patient Name

Examination Room

Super keys : {RoomID}, {RoomID, RoomType}

Candidate key : {RoomID}

Primary key : RoomID

Appointment

Super keys : {AppointmentID}, {DoctorID, AppointmentDate AppointmentTime}

Candidate key : {AppointmentID}, {DoctorID, AppointmentDate AppointmentTime}

Primary key : AppointmentID

Reason : simpler to use than a composite key

FK: PatientID → Patient.PatientID

DoctorID → Doctor.DoctorID

RoomID → ExaminationRoom.RoomID

2.2 Sample data

Doctor (5 records)

DoctorID	DoctorName	Specialty	Phone	YearsExperience
D01	Somchai Sukjai	Cardiology	081-111-1111	15
D02	Suda Rattana	Orthopedics	081-222-2222	10
D03	Anan Wongsa	Pediatrics	081-333-3333	8
D04	Ploy Chaiyasit	Neurology	081-444-4444	12
D05	Krit Boonmee	Dermatology	081-555-5555	5

Patient (5 records)

PatientID	PatientName	Gender	BirthDate	Phone
P01	Nattapong Srisuk	M	1990-05-12	089-111-1111
P02	Waraporn Jindarat	F	1985-08-23	089-222-2222
P03	Chaiwat Petch	M	2000-01-15	089-333-3333
P04	Siriporn Boonrod	F	1975-11-30	089-444-4444
P05	Thanakorn Meesuk	M	1995-03-08	089-555-5555

ExaminationRoom (5 records)

RoomID	RoomType	Floor	Capacity	Status
R01	General	1	2	Available
R02	Cardiac	2	3	Available
R03	Surgery	2	4	Occupied
R04	Pediatric	1	2	Available
R05	Consultation	3	2	Available

Appointment (5 records)

Appointment ID	PatientID	DoctorID	RoomID	Appointment Date	AppointmentTime
A01	P01	D01	R02	2026-08-20	09:00
A02	P02	D02	R01	2026-08-18	10:00
A03	P03	D01	R02	2026-08-20	09:00
A04	P04	D03	R04	2026-08-19	11:00
A05	P05	D04	R03	2026-08-20	14:00

2.3 Business Question

Q1: Find the name and specialty of doctors who specialize in Cardiology.

Ans $\Pi_{DoctorName,\ Specialty}\ (\sigma_{Specialty = \text{"cardiology"}}\ (Doctor))$

Q2: Find the patient name, doctor name, and appointment date for every appointment.

Ans $\Pi_{PatientName,\ DoctorName,\ AppointmentDate}\ (Patient \bowtie Appointment \bowtie Date)$

Q3: Find pairs of appointments that use the same room on the same date but have different AppointmentIDs.

Ans $\Pi_{A.AppointmentID,\ B.AppointmentID,\ A.RoomID,\ A.AppointmentDate}\ (\rho_A(Appointment) \bowtie A.RoomID = B.RoomID \wedge A.AppointmentDate = B.AppointmentDate \wedge A.AppointmentID \neq B.AppointmentID\ (\rho_B(Appointment)))$

Q4: Find the IDs of doctors who never had an appointment.

Ans $\Pi_{DoctorID}\ (Doctor) - \Pi_{DoctorID}\ (Appointment)$

Q5: Find the names of doctors who have an appointment on 2026-04-20 in a room located on the 2nd floor.

Ans $\Pi_{DoctorName}\ (Doctor \bowtie (\sigma_{AppointmentDate = \text{"2026-04-20"}}\ (Appointment) \bowtie \sigma_{Floor=2}\ (ExaminationRoom)))$

Extra: Count the total number of appointment for each doctor.

Ans $DoctorID\ \gamma\ count(AppointmentID)\ (Appointment)$